Joao Victor Quintanilha

> software engineer · ai systems · physical ai & ml research
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01 Education

Tufts University · Medford, MA
M.S. in Data Analytics, concentration: Software Engineering · GPA 3.96 — May 2026
B.S. in Computer Science and Biology — May 2024

- **Awards:** Outstanding Graduate Scholarship Award (top 2 of 1,200 Arts & Sciences graduate students), Honos Civicus, Palantir Tech Scholar, Pegasystems Scholar, SHPE Scholar, ACM Fellow
- **2025 ACM Summer School on Data Science, Greece:** Selected (20/500 globally); graduate coursework in CNNs & Graphs

02 Experience

Dimenso · Cambridge, MA — Feb 2026 – present
Software Engineer / AI Integration Team
Python · FastAPI · PostgreSQL · React · Electron · LiveKit · Claude API · AWS

- Architected a provenance engine attributing AR-glasses and camera captures (photos, video, OCR, tables) to protocol steps via an append-only event log, interval-overlap matching, and EXIF/filename capture-time recovery
- Synced protocol step state across a LiveKit voice agent, AR smart glasses, and a desktop console over WebSockets; the agent narrates each step from LLM-pregenerated summaries
- Hardened Claude-powered generation for 4–8 h lab sessions with streamed output, truncation checks with retry, and priority input budgeting; scaled report fill to 128k tokens, recovering outputs that failed silently
- Extended the AI workspace agent to draft protocols from recorded sessions and fill registry tables from photos of instrument printouts, Word, and Excel files
- Replaced N+1 lineage traversal (tens of thousands of SQL queries per page) with one bounded recursive CTE, loading 2,000-node biological lineage graphs in milliseconds
- Migrated the sample registry to multi-tenant org ownership with role-based access control, and built the import that moved a biotech customer off Benchling with original IDs intact
- Built FastAPI/WebSocket pipelines streaming lab instrument data into AI agents; ML table extraction cut LLM noise by 20%

Tesla · Austin, TX — Sep 2025 – Dec 2025
Software Engineer Intern / Infrastructure Team
React · .NET Core · C# · SQL Server · Splunk

- Built a React + .NET Core developer platform for commissioning workflows used by 200+ engineers and managers
- Improved platform scalability by refactoring 20k+ lines of code into reusable, ARIA-compliant React components
- Enabled analytics on 100k+ records by designing normalized SQL Server schemas with FKs, indexes, and audit trails
- Developed and maintained 27+ REST APIs with batch ingestion, and unit/integration tests with 80%+ coverage

German Aerospace Center (DLR) · Braunschweig, Germany — Jun 2025 – Aug 2025
Software Engineer Intern / Visual Analytics Web Application
PostgreSQL · pgvector · Transformers · React · GitHub Actions

- Increased semantic search accuracy by 35% by building PostgreSQL + pgvector ANN indexes for cosine RAG retrieval
- Enabled multilingual retrieval across 12k+ docs by deploying transformer-based embedding models for semantic search
- Built REST APIs for vector indexing, retrieval, and metadata management; integrated with CI/CD pipeline via GitHub Actions
- Improved analyst productivity by integrating React + MUI dashboards for interactive document exploration

03 Research & Projects

Sperm Whale Bioacoustics Research · Tufts University — 2026
Co-First Author / accepted at DCASE 2026 Workshop (MIT) ↗ paper, code & results
PyTorch · contrastive learning · audio ML

- Proposed DCCE, a contrastive model encoding sperm whale coda rhythm (GRU over inter-click intervals) and spectra (CNN over log-mel) in separate branches before fusion
- Beat frozen bioacoustic foundation models (WhAM, AVES, BirdNET) on coda-type classification, 0.865 vs. 0.722 macro-F1, on 1,383 codas from the Dominica Sperm Whale Project
- Designed recording-date-disjoint evaluation that exposed generalization limits hidden by random splits

Hi!ckathon · Institut Polytechnique de Paris — Nov 2024 – Dec 2024
AI Solutions Developer / AI-Driven Mobile Application ↗ GitHub
XGBoost · LightGBM · FastAPI · iOS

- Won 1st place in Technical Excellence at the Hackathon, for delivering the best model of the event with 69% F1 score
- Trained and integrated a hybrid model (Neural Networks, XGBoost, LightGBM) to predict groundwater levels in France
- Developed a FastAPI to integrate the ML model with an iOS app, enabling real-time predictions through API endpoints

04 Skills

languages	Python, TypeScript, SQL, C/C++, C#, Java	backend	FastAPI, Node.js, .NET Core, REST, WebSockets
data	PostgreSQL, pgvector, SQL Server, MongoDB, Pandas	ai/ml	PyTorch, TensorFlow, Claude & OpenAI APIs, RAG
agents	LiveKit voice agents, LangChain, Claude Code	cloud/devops	AWS, GCP, Docker, GitHub Actions, Linux